› Prepared by Danielle McGuire and Greyfox for the May 2026 session of the Society of Hackers

ADVANCED MESH HACK TRICKS



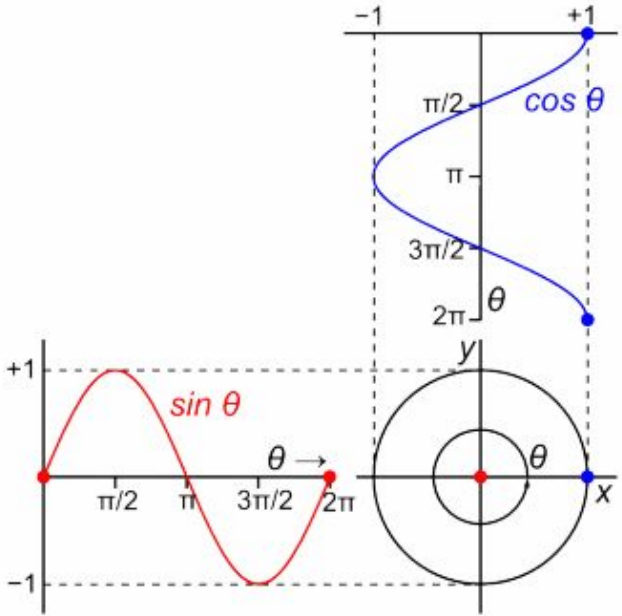
Radio Measurement and Hardware Fabrication

Danielle McGuire

Meshtastic is a platform for encrypted off-grid messaging utilizing the LoRa protocol in an ISM band (US: 915 MHz). What does that even mean?

DEMO - Fundamentals of Radio Waves

If you want to find the secrets of the universe, think in terms of energy, frequency and vibration. Nikola Tesla



Using VCVRack, demonstrate wave amplitude, frequency, wavelength, waveshape, and modulation

If that don't work - https://www.codeitbro.com/tool/sine-wave-generator

LoRa Protocol

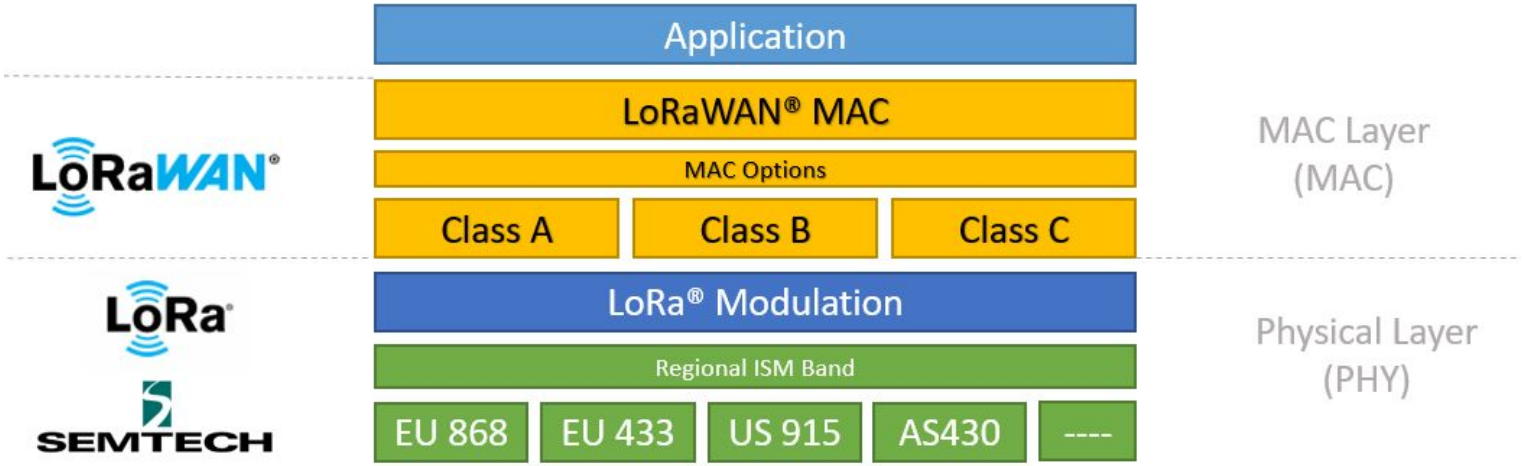


Image Credits: Semtech. Semtech makes the SX1262 chip which Heltec uses on its LoRa boards

- SDR capable of RX/TX over 150-960 MHz, up to +22dBm with integrated power amplifiers

In the diagram above, the Physical Layer corresponds to OSI Layer 1 and the MAC Layer corresponds to OSI Layer 2

- LoRaWAN is a centralized star network for connecting LoRa radios to the Internet (similar to Zigbee)
- *LoRa does not implement device authentication, and neither does Meshtastic.*

DEMO - Dissecting the LoRa Protocol with an RTL-SDR

Postponed for future presentation

Using an RTL-SDR, capture communication between two Meshtastic nodes over the unencrypted LongFast channel, then decode that data using GNU-Radio to recover the payload
https://github.com/tapparelj/gr-lora_sdr

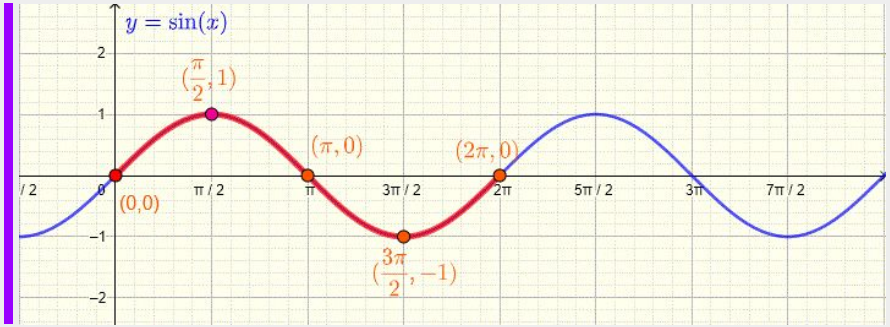
Radio, like all STEM fields, is all about measurement.

The measurements we care about used to require tools that cost $10K-30K total; today, they cost less than $200.

What Are We Measuring?

On a 2D graph, the **domain** is the horizontal x-axis, and the **codomain** is the vertical y-axis, while the **range** is the set of all values the function takes.

Here, the domain is all real numbers, the codomain is all real numbers, and the range is all real numbers between -1 and 1.



COMMON MEASUREMENT DOMAINS:

- Time Domain (Oscilloscope)
- Frequency Domain (TinySA, NanoVNA)

COMMON MEASUREMENTS:

- Signal amplitude (Oscilloscope, TinySA)
- Complex Impedance (NanoVNA)
- Parameter Domains (NanoVNA)
  - S-Parameter Family
    - Standing Wave Ratio (SWR)
  - Z-Parameter Family
- And more! (other equipment)
  - H-, T-, ABCD-Parameter Families

TinySA Measurements - Spectrum Readings

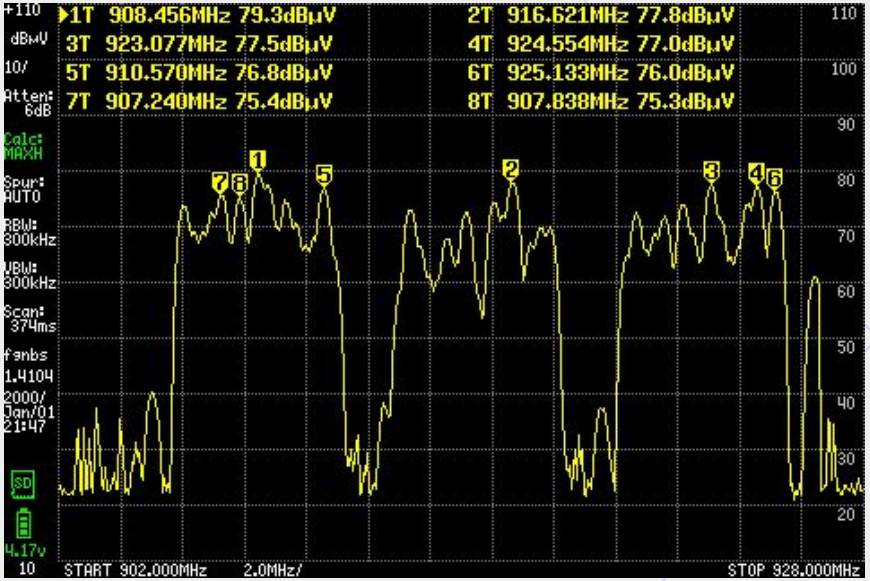
Frequency-domain signal amplitude measurements
- AKA “Frequency on x-axis, Signal Strength on y-”

Lets us “see” the signal as it passes by*
*sampling rate can warp results

$63 standard (0.1-960 MHz), $183 Ultra (0.1-7000 MHz) on AliExpress

Also need protection equipment (SoH provides access to
- 10W attenuator ($25 10W 20db 6G* unit),
  *6G means “DC (0 Hz) to 6 GHz input range”
- Set 2W attenuators ($45 2W 1-20db 8G 6pc set)
- RF filter ($78 VLM-33W-2W-S unit))

Or you can use an $38 RTL-SDR as a 24-1700 MHz Spectrum Analyzer! Check out Paul Maine’s tutorial



From TestMeasurementTips.com: *Radiated emissions in the 915 MHz ISM band at a nearby drug store to the gas station price sign. The analyzer was placed in Max Hold mode, and a short 900 MHz antenna was used. The TinySA Ultra was placed in my shirt pocket while walking throughout the store for a few minutes to build up a composite spectral plot. This corresponded to the interference received at the gas station.*

NanoVNA Measurements

Capable of showing multiple measurement domains at once

- Freq Domain, plotted on Smith Chart
  - “Smith” (green trace) - Complex Impedance
- Freq Domain, plotted on Cartesian axes
  - “LogMag” (yellow trace) - Attenuation
  - “SWR” (blue trace) - Standing Wave Ratio
  - “|Z|” (purple trace) - Impedance Magnitude

We care most about SWR and Impedance matching

- Let’s look at each separately and explain what those big words mean

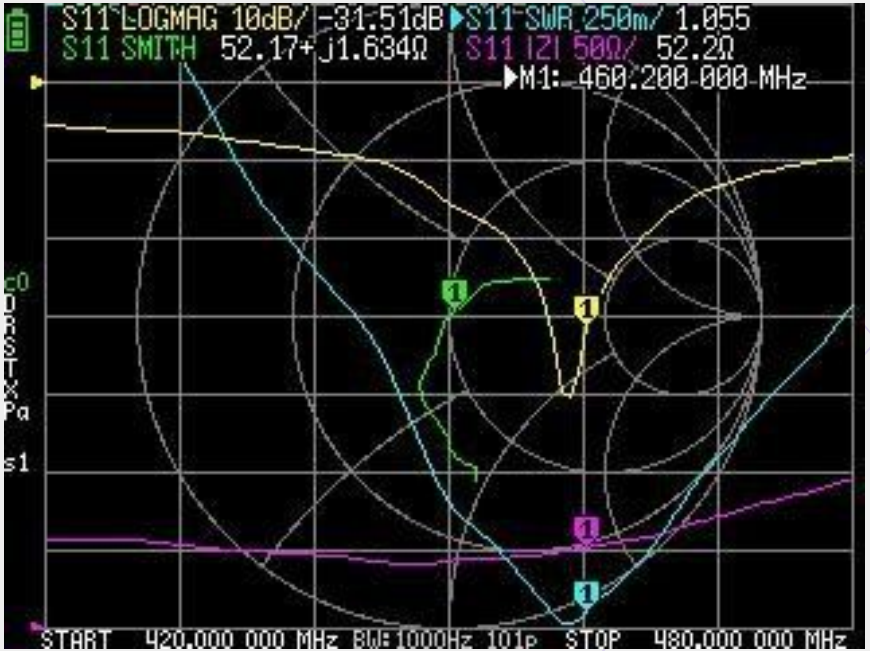


Photo from [dosw’s post on forums.mygmrs.com](forums.mygmrs.com) measuring $75 MXTA26 antenna for GMRS (462-467 MHz) (min SWR at 465 MHz)

NanoVNA Measurements - s-parameters Theory

Top left: diagram of 2-port network

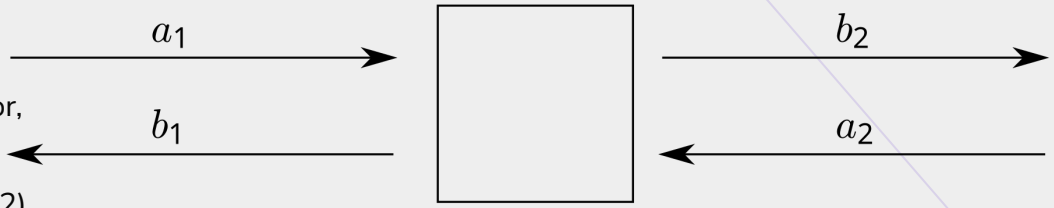
- Component with two wires (resistor, capacitor, diode, antenna, etc)

Middle left: *scattering parameter matrix* equation relating inputs (*a*) to outputs (*b*) at each port (1 and 2)

- Ignore the equations if they make no sense

Bottom left: s_{in} is SWR, *standing wave ratio*

- Also called *voltage standing wave ratio*, or VSWR
- Ratio of max voltage to min voltage on the antenna
  - 1 is perfect
  - <1.1 is excellent for an antenna
  - <1.6 is typical for a <$5 antenna
  - >=2 is a terrible antenna



$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

$$s_{\text{in}} = \frac{1 + |S_{11}|}{1 - |S_{11}|}$$

Photo credit: Wikipedia

NanoVNA Measurements - SWR Matching

Just need to know the SWR value at desired freq, and the freq at which SWR is minimum

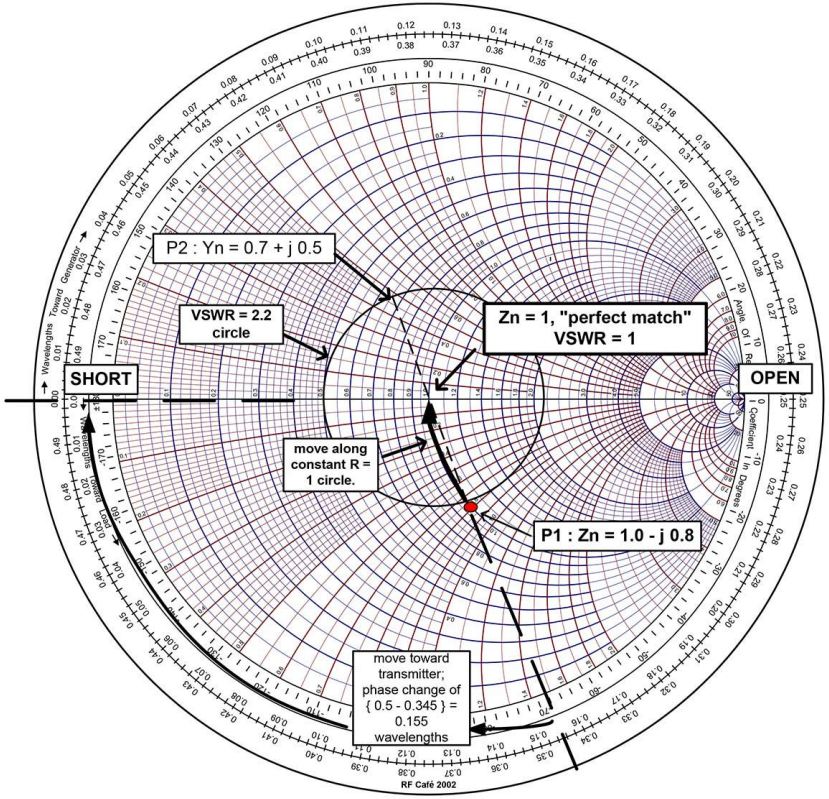
- Can tune SWR on some antennas by clipping off small pieces until min SWR freq is desired RX/TX freq

In other words - just read a graph!



Photo credit: [Kerry Wong, "Teardown of a LiteVNA Vector Network Analyzer"] (The LiteVNA project forked off NanoVNA, this screen looks the same on NanoVNA)

NanoVNA Measurements - Smith Chart Theory



Smith Chart - alternative to Cartesian graph (x and y axis) for plotting complex impedance values over freq domain

- More counterintuitive at first, but turns tricky algebraic calculations into easy geometric observations
  - Far left on center horizontal axis is *short* (zero imp.)
  - Far right is *open* (infinite imp.)
  - Dead center is 50ohm (radio standard)
- Going up adds capacitance (positive imaginary)
- Going down adds inductance (negative imaginary)

When impedance matching antenna, want impedance graph to be physically closest to center at operating freq

- Alternate side of circle (see P2 compared to P1) gives impedance to add to antenna circuit to get to 50ohm total

Photo credit: Jude Misaid

NanoVNA Measurements - Reading the Full Graph

Look at each line separately and understand its domain

- Same freq at same marker (we have 1 here)
- Smith chart is separate from Cartesian axes

Remember:

- Need to calibrate for each measurement range (can save 5-7 calibrations depending on model)
- All these values change based on antenna position/orientation, surrounding objects, and atmospheric conditions

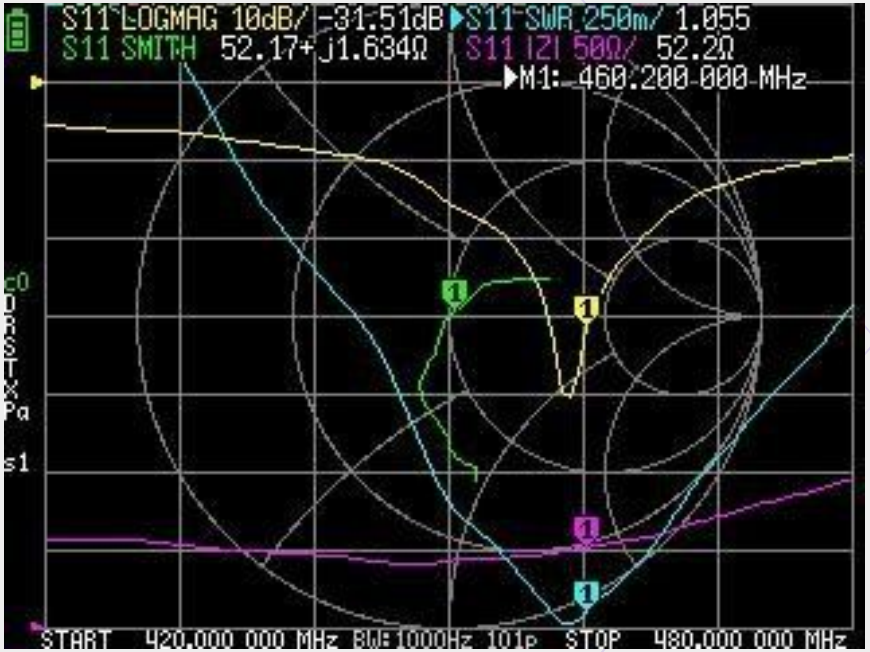
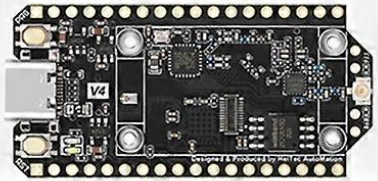


Photo from [dosw's post on forums.mygmrs.com](forums.mygmrs.com) measuring $75 MXTA26 antenna for GMRS (462-467 MHz) (min SWR at 465 MHz)

LoRa Mesh Nodes need 4 parts:
LoRa Chipsets,
LoRa Antennas,
Power Supply, and
Cases





Parameters	V3	V4
MCU	ESP32-S3FN8	ESP32-S3R2
Flash	8MB (integrated)	16MB (external)
Measured Distance Reference	10-12 km	25-30 km
Power Management	Standard Li-ion management IC	Upgraded IC supporting solar input
Solar Input LoRa	⊗	Added SH1.25-2P interface
TX Power	21±1dBm	28±1dBm
2.4G Antenna	Metal spring antenna	FPC antenna + IP EX connector
GPS Interface	⊗	Added SH1.25-8Pin interface
Peripheral I/F	36-pin	40-pin
Screen Connection	Soldered	Soldered + B2B connector
Pin Craft	Silver-plated	Gold-plated
Screen Bracket	Partial protection	Full protection

Chipset - Heltec ESP32

V3 ($18)- **ESP32S3-FN8 MCU** (Wifi, Bluetooth) and Semtech **SX1262 LoRa Node chip** (LoRa), with USB-C and **CP2102 USB-to-serial chip**

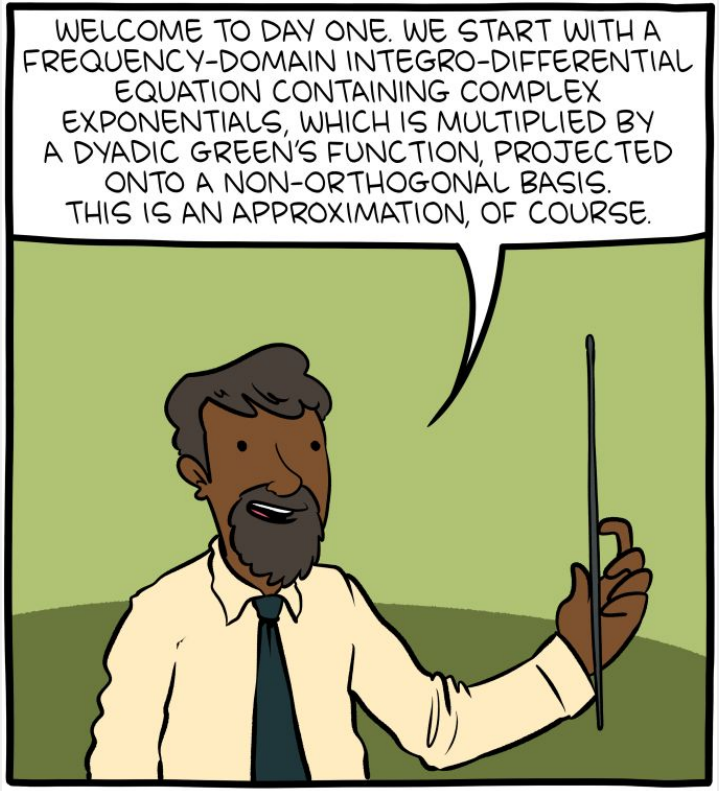
V4 ($20) - All that, but upgraded **ESP32S3-S3R2 MCU, 4x TX power, built-in solar charge controller,** and **much nicer GUI**, with identical pinout. S3R2 USB-OTG feature means no need for C2102, lowering power consumption

Having Wifi Access and a GUI screen is very convenient for end users, but burns the battery much faster
Ideal for handheld mobile nodes

Antennas - Build Your Own!

Easy - Steven's Stinger Gen2
Tricky - Steven's Stinger Gen3
Eldritch and Abominable - Design Your Own
have fun doing ANTENNA MATH, dummy

To make a decent antenna, *very carefully* cut pieces to the length given in a design, or determined by a good antenna calculator
To make a great antenna, measure that with a VNA, and snip tiny pieces off of elements until resonant frequency and impedance are as perfect as possible



Everyone hates on reductionism in physics until they see what engineers have to do to describe a simple antenna.

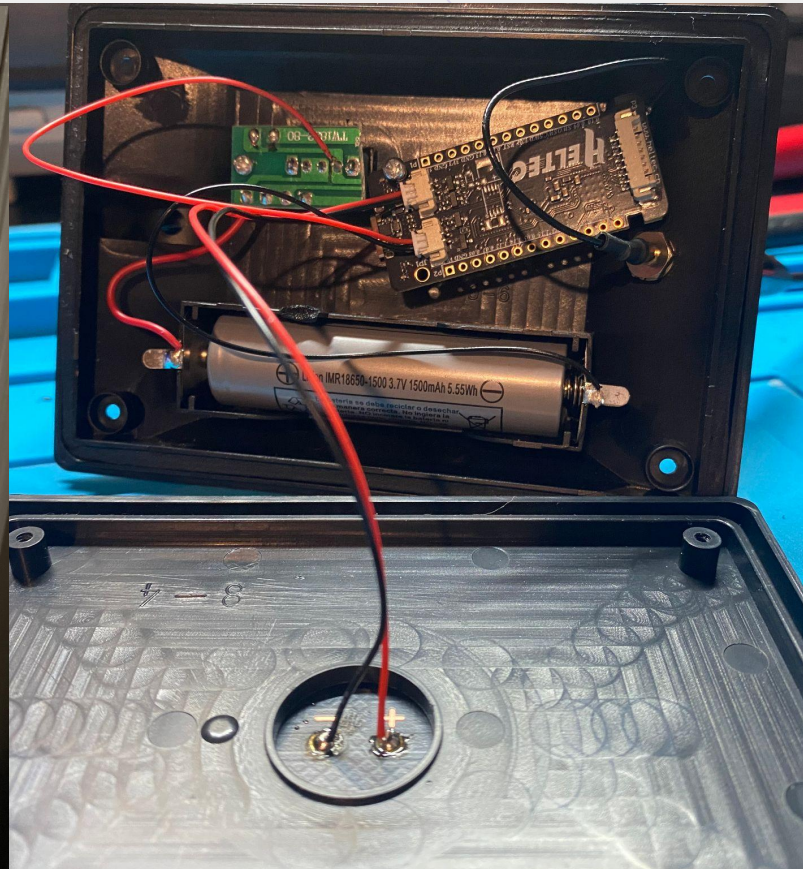
Power Supply and Case - Stationary Solar Node

HACKERSBLOOMv1
$30 components

- Lowe's 1W Harbor Breeze solar light
- Heltec T114
- GT-800 antenna

MUST be positioned where solar panel gets max sun!!
Antennas should be vertical for best reception





Power Supply and Case - Handheld Mobile Node

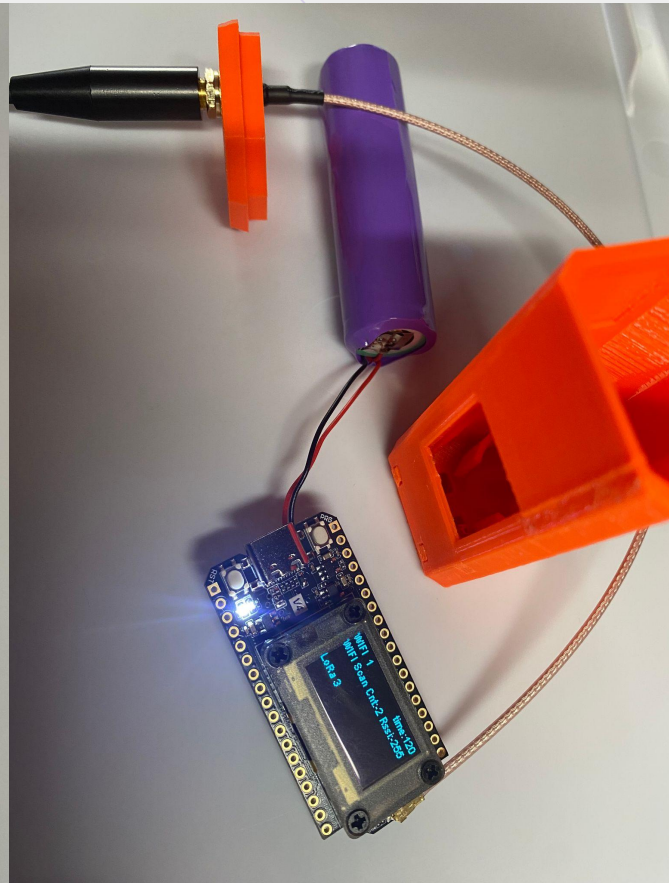
NOMADv1
$30 components

- Heltec v4
- GT-800 antenna
- Battery pack
- PETG case

Currently researching:

- Building cheaper, better antennas at scale
  - Hinged? (best performance when antenna vertical)
- FakeTec boards
- Lid pops off case in current model





Questions?

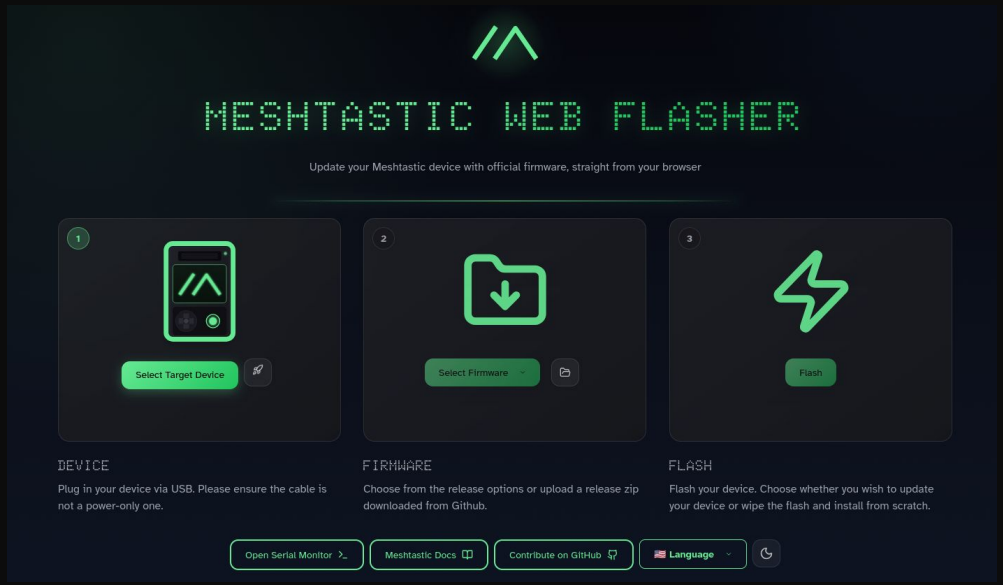
Meshtastic Configuration



Grey Fox

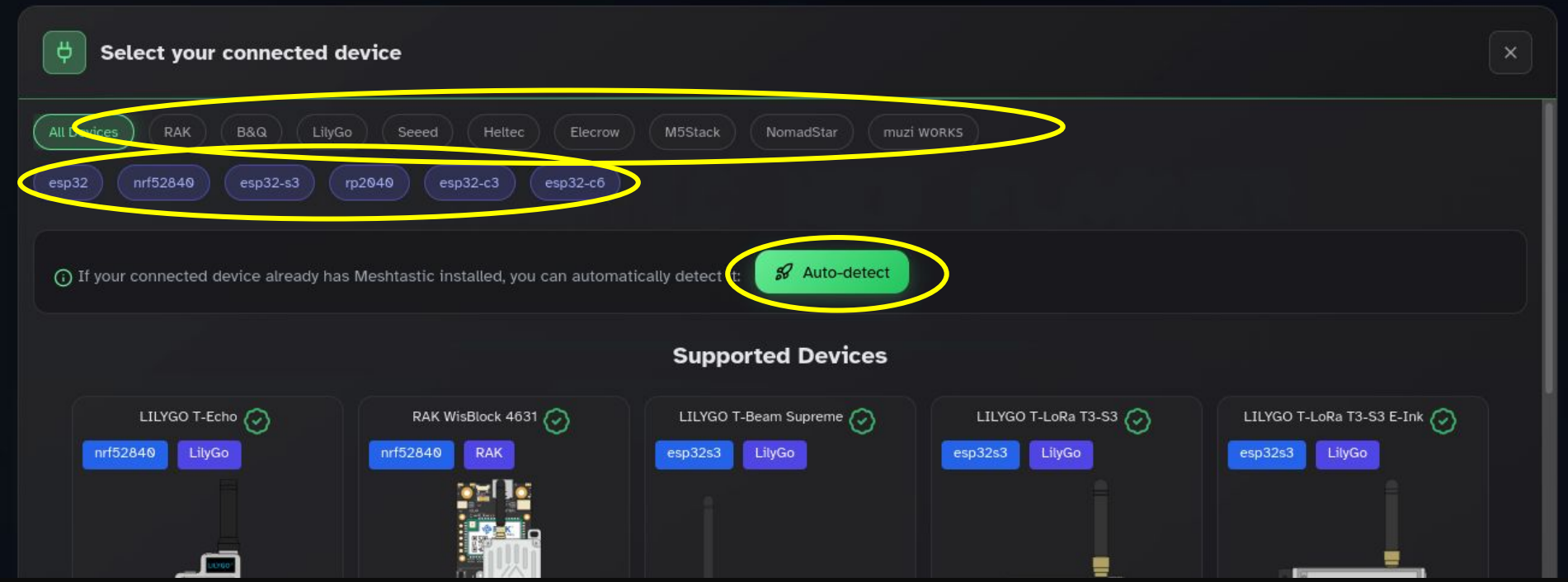
Flashing Firmware

https://flasher.meshtastic.org/

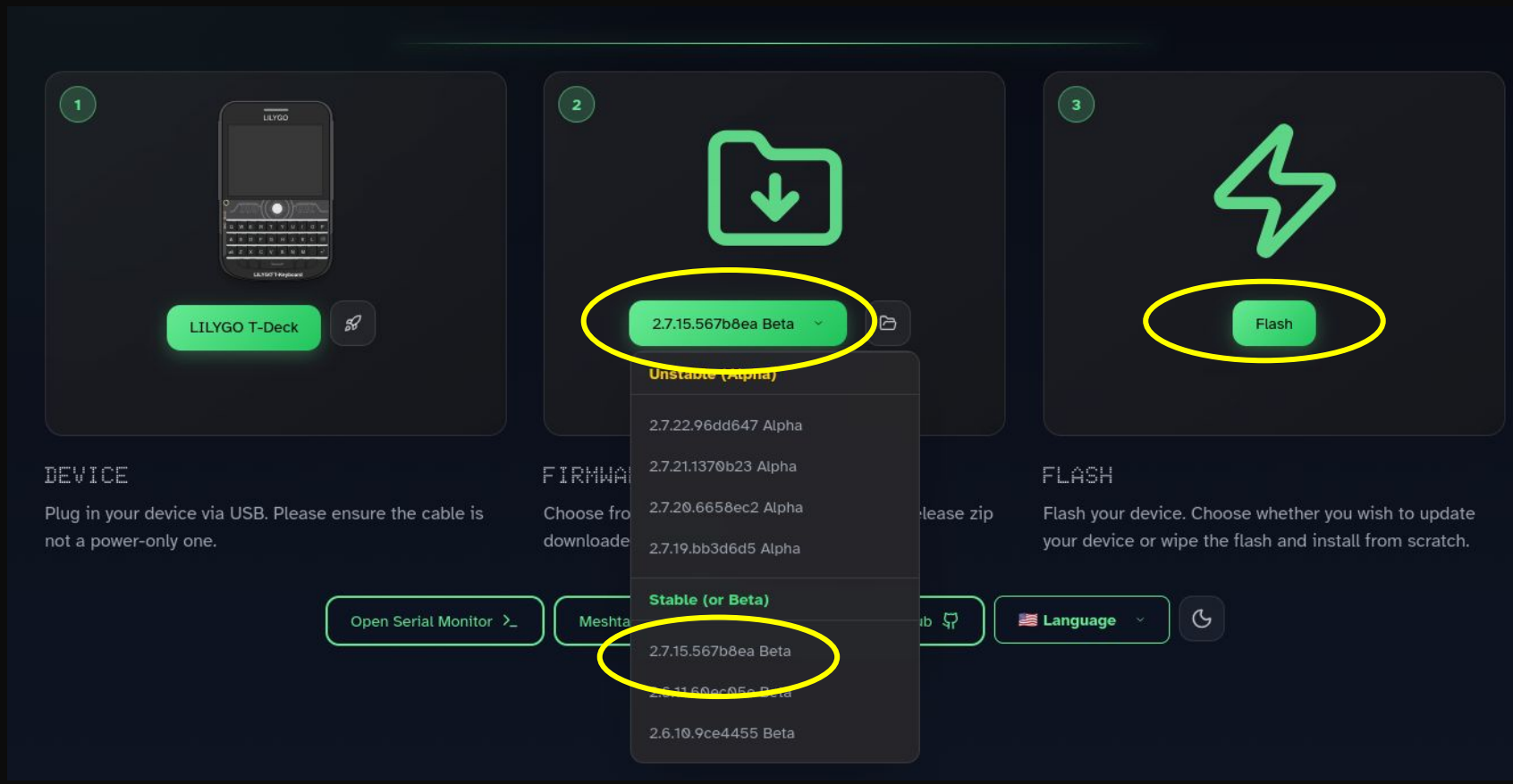


Requires Chromium-based browser

Selecting Target

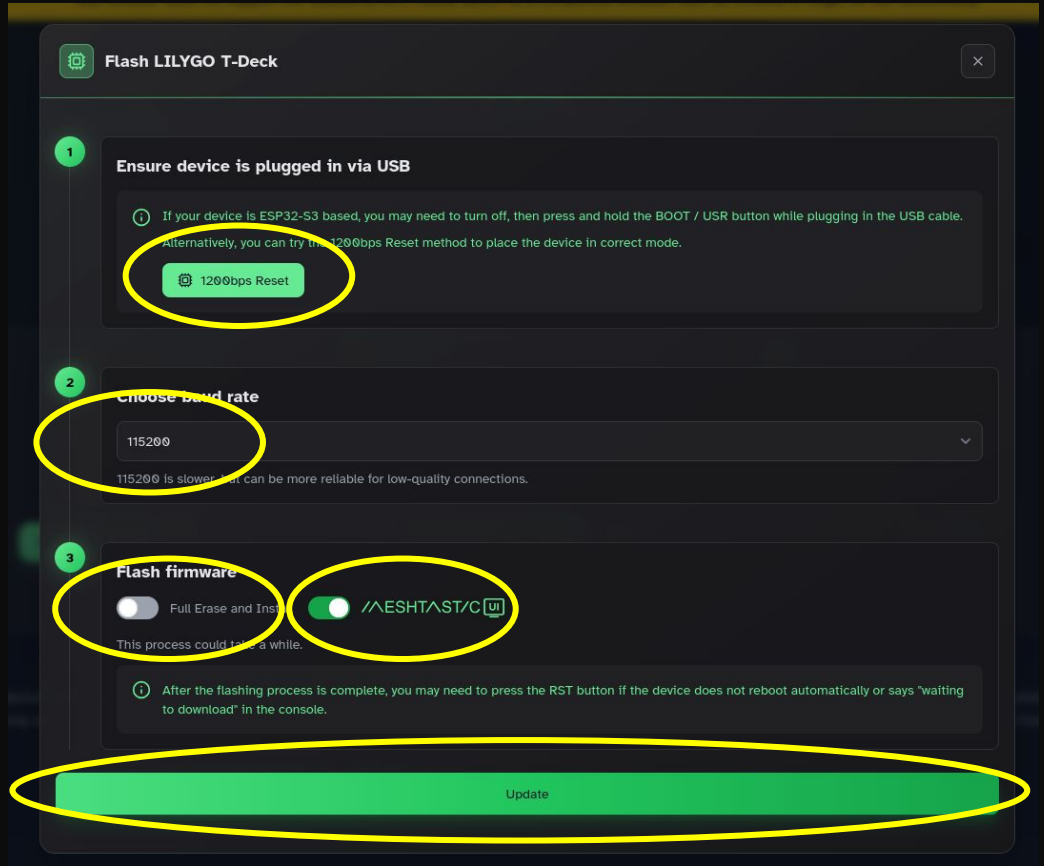


Firmware Version



Firmware Version

- Boot mode is not the same for all devices
- nRF62840 usually auto-switches to bootloader
- Some devices require manual file upload
- USB cable is likely the issue
- Some GNU/Linux distros won't play ball



Baseline Configuration

- **Meshtastic mobile app + BLE tether**
- **LoRa**
- **Region:** United States
- **Presets:** LONG_FAST
- **Number of Hops:** 6 (depends on node density)
- **User**
- **Long Name:** dealer's choice
- **Short Name:** 4 characters
- **Licensed Amateur:** HAM operators have a reserved space — no encryption allowed

Device Role

- Device
  - Device Role: CLIENT
  - Rebroadcast Mode: ALL

Client	Send and receive all
Client Mute	No routing or repeating
Client Base	Favorite nodes get priority—rooftop choice
Client Hidden	Send and receive only user-designated channels
Router/Repeater	Always rebroadcasts, never originates, no deferment
Router Late	Like router, but defers to concurrent routing
Sensor	Telemetry over text messaging
Tracker	Tracking tag

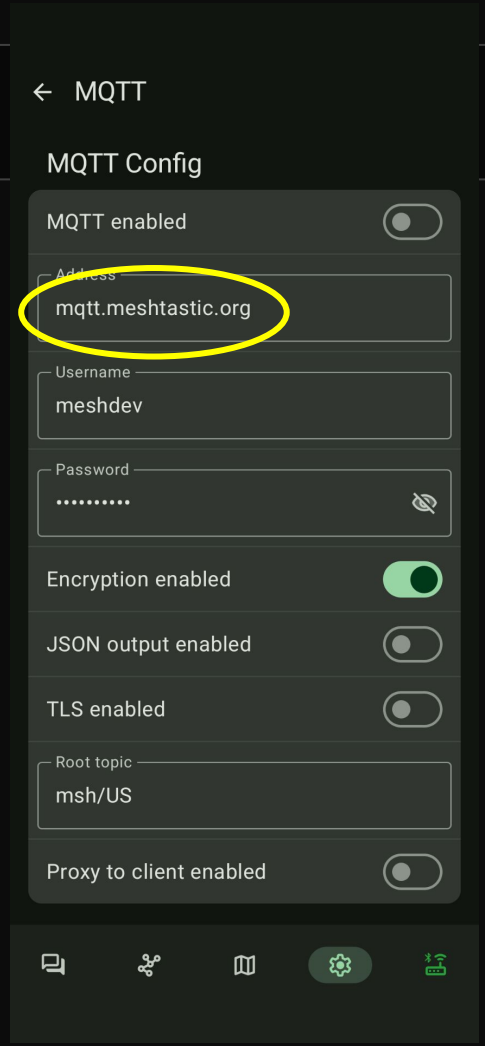
More Configuration



- **Position:** user's choice
- **Power:** important for Sensor, Tracker
- **Display:** nice-to-have options; keep default
- **Bluetooth:** no-screen devices have default PINs
  - 0000
  - 1234
  - 123456
- **Consider changing them**

Module Configuration

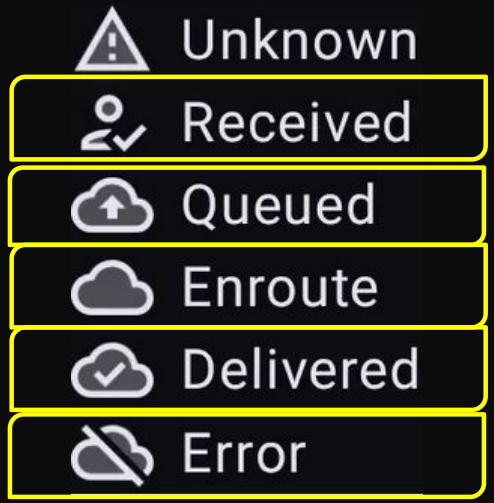
- **MQTT:** it depends
  - **Default:** https://mqtt.meshtastic.org/
  - **Consider:** https://mqtt.wpamesh.net/
- **Serial:** computer tethering options
- **External Notification:** LED, buzz, and vibrate
- **Store & Forward:** if acting as a server, enables infrastructure for message control
  - Only ESP32 based devices with onboard PSRAM like the T-Beam and T3-S3 can be a Store & Forward Server.



Useful App Stuff

- **Nodes Tab**
  - Mark favourites
  - Check hardware
  - Send a direct message
- **Mesh Map:** only if GPS enabled
- **NodeDB Reset:** empties recorded nodes, starts over
- **Traceroute:** doesn’t always work, but gives good awareness of routing and hops

Messaging



Direct message received and acknowledged

Waiting for your node to connect

Waiting for a node receive/ack it

Message received by mesh

No network or hop/rebroadcast limit exceeded

Channels

- **Settings > Channels**
- Key auto-generated based on name
- Limit 7 channels
- LongFast is default primary channel
  - Changing Primary forces manual frequency slot configuration
- Uplink/Downlink: allows send/receive from public internet gateway
  - Gateway configured in MQTT settings
- Position: shares location if active

Questions?

THANK YOU!

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